M^{3}RIS: Mutual Modulation Mamba Network for Referring Remote Sensing Image Segmentation

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***Abstract*—Referring remote sensing image segmentation (RRSIS) aims to map natural language queries to pixel-level masks within complex geographic scenes, requiring precise cross-modal semantic alignment. In recent years, Mamba has demonstrated impressive performance in various remote sensing tasks due to its powerful contextual modeling capability, offering new potential for RRSIS. However, vanilla Mamba primarily models sequential dependencies and provides limited support for cross-modal interaction, thus hindering precise vision–language semantic alignment. To overcome this limitation, we propose M^{3}referring image segmentation (RIS), a mutual modulation Mamba network for RRSIS. Specifically, we propose the vision–language mutual modulation (VLMM) module that distills semantic knowledge (SK) from the segment anything model (SAM) to provide auxiliary semantic cues for cross-modal interaction. The module not only uses the distilled knowledge to guide the prediction of channel-wise affine coefficients and spatial dynamic convolution kernels to enhance target-relevant visual responses, but also introduces this knowledge into cross-attention as a semantic bias to enable language features to aggregate semantically relevant visual context. This facilitates mutual enhancement between vision and language, promoting more effective cross-modal alignment than conventional unidirectional modulation. Moreover, we introduce the hybrid semantic refined (HSR) decoder, which alternates between visual state space (VSS) blocks and cross-modal semantic refinement (CSR) modules. The VSS blocks capture global semantics via long-range context modeling, while the CSR modules derive multiscale dynamic convolution kernels from language features to refine local visual features through dynamic filtering, thereby enhancing semantic awareness of the referred object. Extensive experiments on RRSIS-D, RISBench, and RefSegRS datasets demonstrate that M^{3}RIS outperforms several previous state-of-the-art approaches.**

***Index Terms*—Cross-modal semantic alignment, Mamba, mutual modulation, referring remote sensing image segmentation (RRSIS).**

I. Introduction

REFERRING remote sensing image segmentation (RRSIS) is an emerging yet frontier task in visual-language multimodal understanding, aiming to precisely segment objects in remote sensing images based on natural language descriptions. Compared with conventional unimodal remote sensing semantic segmentation [1], [2], [3], RRSIS facilitates more precise interactive object localization. Its segmentation categories are entirely determined by natural language expressions, enabling nonexpert users to accurately retrieve specific targets from remote sensing images. This substantially enhances the efficiency of image interpretation and holds broad application value and prospects in fields such as environmental monitoring [4], land use planning [5], urban management [6], and disaster emergency response [7].

RRSIS remains a promising direction within vision–language multimodal analysis for remote sensing, demanding continuous methodological advancements. Given the absence of task-specific datasets during its infancy, Yuan et al. [8] construct RefSegRS, the first specialized dataset for referring image segmentation (RIS) in remote sensing, thereby pioneering the RRSIS task. They also introduce a language-modulated cross-scale transformer architecture for improved localization and segmentation of small and dispersed objects in complex remote sensing scenes. However, RefSegRS contains only 4420 image-text pairs, which is insufficient for complex semantic reasoning or generalization to real-world scenarios. Given insufficient data scale and diversity, RMSIN [9] proposes RRSIS-D, which expands the benchmark scale by nearly fourfold and supports more challenging scene understanding. Further advancing fine-grained cross-modal alignment, Lei et al. [40] parse each referring expression into two components, namely the described ground object and its associated spatial context, to finely modulate visual features. While prior methods apply language modulation only during encoding or fusion, recent works such as CADFormer [10] and LSCF [11] extend it to decoding, allowing language to exert stronger modulation over segmentation.

Most of the aforementioned methods adopt transformer architectures and generally follow a unidirectional language-to-vision (L2V) modulation paradigm. Under this paradigm, only visual features are modulated based on language features, whereas language representations lack visual feedback, limiting effective cross-modal interaction. In recent years, Mamba [12] has attracted extensive attention, as it generally models long-range dependencies more efficiently than transformers [13]. With this advantage, Mamba has demonstrated remarkable performance in various remote sensing tasks, such as change detection [14], semantic segmentation [15], and object detection [16], offering new potential for

Received 15 February 2026; revised 8 May 2026; accepted 1 September 2026. Date of publication 7 September 2026; date of current version 21 September 2026. This work was supported in part by the National Natural Science Fund of China under Grant 62221002 and Grant U24A20327 and in part by Hunan Provincial under Grant 2026JJ20061 and Grant 2018TP1013. *(Corresponding author: Bin Sun.)*

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Digital Object Identifier 10.1109/TGRS.2026.3730817

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RRSIS. However, when directly applied to RRSIS, vanilla Mamba still falls short of supporting robust vision–language interaction, thereby impeding precise cross-modal semantic alignment.

To tackle this challenge, we introduce the mutual modulation Mamba network for RRSIS, dubbed M^{3}RIS. Distinct from previous paradigms, M^{3}RIS is built upon Mamba and introduces bidirectional vision–language modulation, achieving more effective cross-modal alignment than conventional unidirectional modulation. Meanwhile, semantic knowledge (SK) distilled from the segment anything model (SAM) [17] provides auxiliary semantic information to further improve alignment. The core components of our method are the vision–language mutual modulation (VLMM) module and the hybrid semantic refined (HSR) decoder. The VLMM module distills SK from an off-the-shelf SAM into a lightweight predictor, eliminating the need for SAM during inference. In the L2V branch, the distilled SK guides the prediction of channel-wise affine coefficients and spatial dynamic convolution kernels, enhancing target-related visual responses. In the vision-to-language (V2L) branch, this distilled knowledge is introduced into cross-attention as a semantic bias, guiding language representations to aggregate semantically relevant visual context. This mutual modulation promotes cross-modal semantic alignment while mitigating the limited multimodal interaction of vanilla Mamba. The HSR decoder alternates between visual state space (VSS) blocks and cross-modal semantic refinement (CSR) modules, integrating global contextual modeling with language-guided local refinement. Here, VSS blocks capture long-range context for global understanding, while CSR modules derive multiscale dynamic convolution kernels from VLMM-modulated language representations to refine local visual features through dynamic filtering, thereby enhancing semantic awareness of the referred object during decoding.

In brief, the core contributions of this work are outlined as follows.

1) The proposed VLMM facilitates bidirectional cross-modal interaction by distilling SK from SAM. The distilled SK not only guides the prediction of channel-wise affine coefficients and spatial dynamic convolution kernels to enhance target-relevant visual responses, but also serves as a semantic bias in cross-attention to enable language features to aggregate semantically relevant visual context, thereby achieving more effective cross-modal semantic alignment than conventional unidirectional modulation.
2) The introduced HSR decoder alternates between VSS blocks and CSR modules, where the VSS blocks model long-range contextual dependencies and the CSR modules derive multiscale dynamic convolution kernels from language features to refine local visual features through dynamic filtering, thereby enhancing semantic awareness of the referred object.
3) This article presents M^{3}RIS, a Mamba-based network for RRSIS, which achieves state-of-the-art performance on the RRSIS-D, RISBench, and RefSegRS datasets.

II. Related Work

A. Mamba-Based Vision Models

Mamba [12] rivals transformers via selective state-space modeling and hardware-aware design, excelling in NLP and rapidly gaining traction in computer vision. Building on this momentum, vision Mamba [18] pioneers Mamba-based visual architectures through bidirectional scanning for global image context capture. Liu et al. [19] generalize the scan pattern to four directions, thereby improving 2-D spatial dependency modeling. As Mamba-based visual architectures deliver promising results, the vision community increasingly explores their potential for visual modeling. For example, Yao et al. [20] introduce selective visual prompting for Vision Mamba, leveraging lightweight prompts to improve gated feature learning. In parallel, DefMamba [21] presents deformable scanning to dynamically focus on critical regions and enhance structural perception. To improve efficiency, Shen et al. [22] merge similar tokens through cross-layer fusion, reducing training time and peak memory usage. Mimicking retinal ganglion cells, SparX [23] enables dynamic feature aggregation and reuse through sparse cross-layer connections.

The success of Mamba in natural image modeling drives its adoption in remote sensing [15], [16], [24], [25], [26], [27], with change detection emerging as the most prominent application. For instance, Zhao et al. [24] pioneer state-space models (SSMs) in remote sensing change detection, using an omnidirectional selective scanning module for global context modeling in high-resolution images. Additionally, Mamba architectures are also adopted in remote sensing object detection [16], super-resolution [25], classification [26], and semantic segmentation [15] for efficient visual modeling. For instance, Xiao et al. [25] propose a frequency-assisted Mamba framework that improves remote sensing image reconstruction through frequency selection and synergistic local–global modeling, with reduced computational complexity. Cao et al. [26] leverage semantic priors from CLIP and the Mamba architecture to efficiently capture semantic correlations among multisource remote sensing data, improving classification performance and efficiency. Most existing approaches focus on unimodal remote sensing tasks, whereas this work explores Mamba for multimodal RRSIS.

B. Referring Image Segmentation

The RIS task involves segmenting the object referred to by a language expression. Hu et al. [28] first proposed this task, using CNNs and LSTM to process visual and linguistic inputs, respectively, and fuse them through concatenation for pixel-level mapping. Subsequent early works [29], [30], [31], [32] generally follow this paradigm, employing CNNs for visual feature extraction and LSTMs, GRUs, or SRUs for text feature extraction. Among them, [29], [30] experiment with word embeddings for text representation in pursuit of fine-grained vision–language interaction, rather than relying on sentence-level features as in [31] and [32]. Driven by the success of transformers in vision, recent RIS methods [33], [34], [35], [36], [37], [38] increasingly adopt transformer architectures. For example, CGformer [33] leverages learnable query tokens

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and contrastive learning in a grouped transformer to enable object-level cross-modal reasoning. Recently, to enable more efficient multimodal reasoning, ReMamber [39] introduces the Mamba architecture to the RIS task for the first time. Yet, current Mamba-based RIS research remains in its nascent stage, calling for further exploration and advancement.

Continual progress in RIS for natural images has inspired research in remote sensing. LGCE [8] pioneers the RRSIS task and introduces the RefSegRS dataset, together with a language-guided cross-scale enhancement module targeting the segmentation of small and scattered objects. Subsequently, RMSIN [9] constructs the larger RRSIS-D dataset and employs intrascale and cross-scale interactions, along with adaptive rotational convolution, to tackle challenges posed by complex spatial scales and rotation variations in remote sensing images. Since then, research interest in RRSIS has continued to grow, with a series of advances [10], [11], [40], [41], [42], [43] driving progress in the field. For instance, Lei et al. [40] decompose referring expressions into ground objects and spatial relations, extracting fine-grained linguistic features separately to enhance image–text alignment. DANet [42] uses dual alignment to reduce cross-modal domain shifts and suppress semantically irrelevant predictions in implicit optimization. Diverging from prevailing paradigms, this work proposes a Mamba-based network with bidirectional vision–language modulation, demonstrating its effectiveness in RRSIS.

C. Foundation Models for Remote Sensing Segmentation

In recent years, SAM has attracted growing attention in remote sensing segmentation, with researchers actively exploring its applications in this field. Yan et al. [44] extend SAM to multimodal remote sensing imagery and investigate its applicability across different remote sensing modalities, while Osco et al. [45] explore its use in remote sensing under zero-shot and one-shot settings. Recent studies further adapt SAM to specific remote sensing tasks. FlexiSAM [46] integrates SAM with multimodal fusion to leverage complementary information for high-resolution land-cover classification. ROSSAM [47] fine-tunes SAM with low-rank adaptation (LoRA) for high-quality interactive segmentation of remote sensing moving objects while preserving its generalization ability. SAM is also used to provide object and boundary information for downstream segmentation. Ma et al. [48] incorporate object and boundary cues from SAM into remote sensing semantic segmentation to improve consistency and boundary quality, and Yao et al. [49] explore promptable segmentation for Earth observation. Overall, SAM contributes to remote sensing segmentation not only by generating mask candidates but also by offering complementary cues for object regions and boundaries [50]. This observation also motivates our use of knowledge distilled from SAM in this work.

Recent studies have increasingly explored vision–language foundation models in remote sensing [51], [52], [53], thereby advancing language-guided segmentation. For open-vocabulary remote sensing semantic segmentation, TPOV-Seg [54] uses text-enhanced prompt tuning to exploit pretrained vision–language features and improve generalization to unseen categories, while SegEarth-OV [55] adapts a CLIP-based segmentation framework for remote sensing imagery to support training-free open-vocabulary segmentation. Furthermore, foundation models have been applied to RRSIS. RSRefSeg [56] combines CLIP-based semantic filtering with SAM for mask refinement. RS2-SAM2 [57] improves target localization and boundary delineation through hierarchical vision–text alignment and prompt generation, and DiffRIS [58] leverages a pretrained text-to-image diffusion model to strengthen semantic alignment. Overall, these studies show that foundation models can improve the grounding of referring expressions to target regions and support more accurate mask refinement in RRSIS. Building on these advances, this article incorporates SK distilled from SAM into bidirectional vision–language interaction to enhance cross-modal alignment and improve referring segmentation performance.

III. Methods

Fig. 1 provides a schematic overview of the proposed method. Given a referring expression and its corresponding remote sensing image, the language encoder first extracts linguistic features from the expression. The linguistic features and the image are fed into several mutually modulated encoding layers to facilitate bidirectional vision–language interaction, yielding modulated visual and linguistic features. These are then passed to the HSR decoder, which integrates global contextual modeling with language-guided local refinement to produce the final pixel-wise segmentation mask.

A. Preliminaries

SSM frames sequence modeling as a latent dynamical system, in which an input sequence $x(t) \in \mathbb{R}$ is mapped to an output sequence $y(t) \in \mathbb{R}$ via a hidden state $h(t) \in \mathbb{R}^N$ that evolves over time. The state is updated by integrating incoming inputs through learnable transition dynamics, enabling the model to capture long-range dependencies through a compact representation of past context. The process is mathematically expressed as

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B} \in \mathbb{R}^{N\times 1}$, and $\mathbf{C} \in \mathbb{R}^{1\times N}$ are learnable parameters: $\mathbf{A}$ governs the evolution of the hidden state, $\mathbf{B}$ integrates the input signal into the state, and $\mathbf{C}$ projects the hidden state to the output sequence. To accommodate discrete sequences such as text and images, a time step Δt is introduced, and the system is discretized via the zero-order hold (ZOH) method, resulting in the discrete-time counterparts $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$. Specifically, these are given by

$$\begin{aligned} \overline{\mathbf{A}} &= \exp(\Delta t \cdot \mathbf{A}) \\ \overline{\mathbf{B}} &= (\Delta t \cdot \mathbf{A})^{-1}\left(\exp(\Delta t \cdot \mathbf{A}) - \mathbf{I}\right)\mathbf{B}. \end{aligned} \tag{2}$$

As a result, the system is described by the following discrete-time equations:

$$\begin{aligned} h_t &= \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t. \end{aligned} \tag{3}$$

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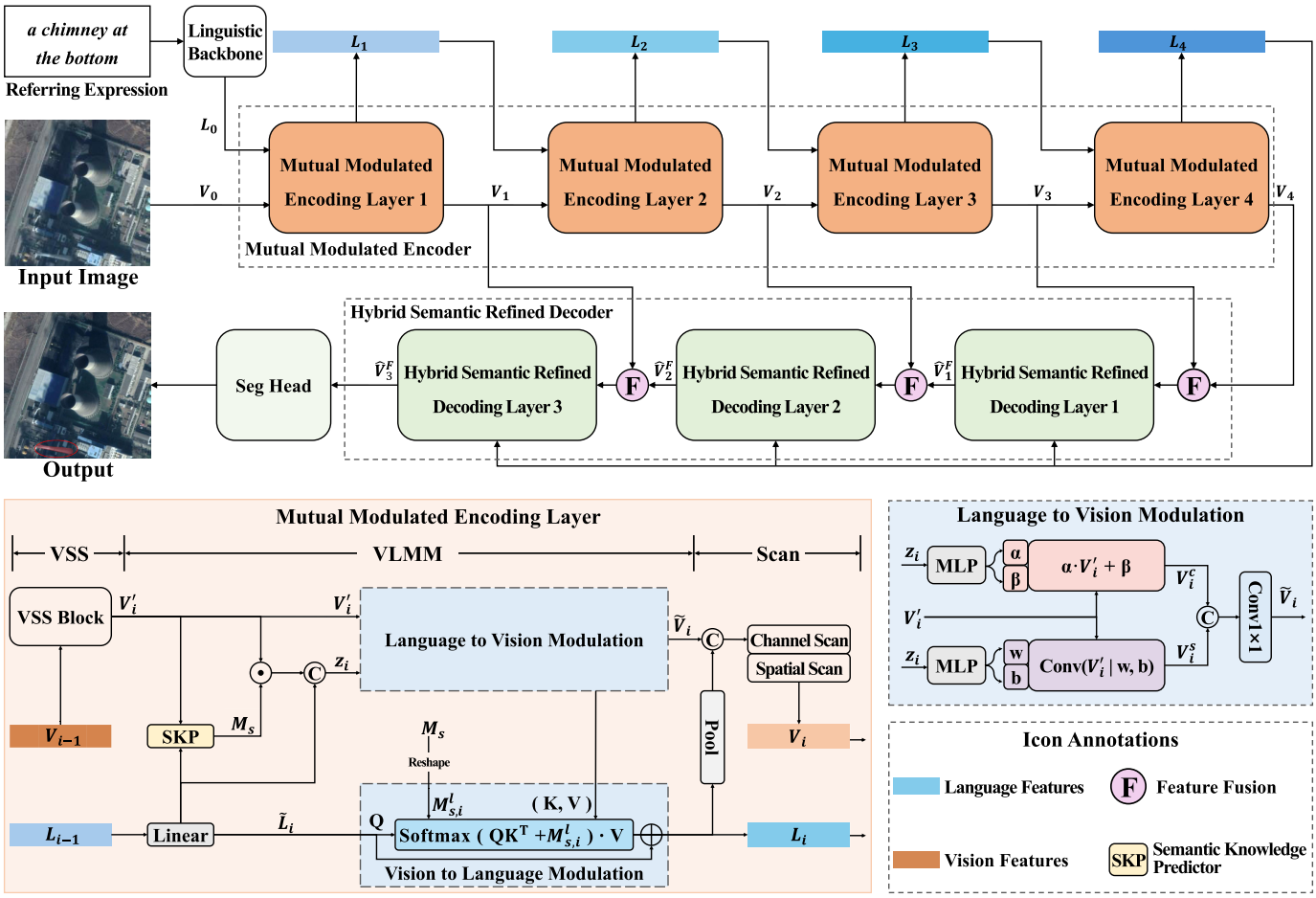


Fig. 1. Overview of the proposed M^{3}RIS method. The mutual modulated encoder consists of multiple mutual modulated encoding layers, each comprising a VSS block, a VLMM module, and a scan mechanism, with the VSS block following the design in VMamba [19]. A lightweight SKP is introduced only in the first-layer VLMM to distill SK from SAM, and the distilled SK is then reused by the VLMMs in subsequent layers to guide vision–language interaction.

In contrast to standard SSMs with fixed parameters, Mamba [12] computes the matrices **B** and **C** in an input-dependent manner through a learned function, enabling selective response to salient information. VMamba [19] further introduces the 2-D selective scan (SS2D), extending this dynamic selection to the visual domain and effectively modeling long-range spatial dependencies in images while preserving linear computational complexity.

B. Vision–Language Mutual Modulation

The design of the VLMM module is presented in Fig. 1. In each mutual modulated encoding layer, the VSS block first extracts visual features $V_i' \in \mathbb{R}^{H_i \times W_i \times C_i}$, which are then fed into the VLMM module together with the language features $L_{i-1} \in \mathbb{R}^{N \times C_{i-1}^L}$ for mutual modulation between vision and language. Notably, when $i = 1$, L_0 corresponds to the original language features L output by the language encoder.

To eliminate the computational overhead of SAM during inference while effectively inheriting its semantic information, we adopt an offline teacher–online student distillation strategy. In the teacher phase, the ground-truth mask is converted to a bounding box prompt b_{box} and fed into the frozen SAM together with the corresponding image I to obtain the teacher SK M_t

$$M_t = \mathrm{SAM}(I, b_{box}). \tag{4}$$

In the student phase, a two-layer convolutional SK predictor (SKP) in the first layer generates the student SK M_s from visual and language features

$$\begin{aligned} L_i' &= \mathrm{Pool}(\mathrm{Linear}(L_{i-1})) \\ M_s &= \mathrm{Sigmoid}\left(\mathrm{Conv}_{1\times1}\left(\mathrm{ReLU}\left(\mathrm{Conv}_{3\times3}\left(\mathrm{concat}\left(V_i', L_i'\right)\right)\right)\right)\right) \end{aligned} \tag{5}$$

with M_t serving as the supervisory signal to transfer knowledge to M_s via a distillation loss. The resulting student SK M_s is reused in subsequent layers without additional SKP computation. It is shared by both the L2V and V2L branches, guiding the prediction of channel-wise affine coefficients and spatial dynamic convolution kernels for visual modulation and serving as a semantic bias in cross-attention for language refinement. The detailed formulation of the distillation loss is provided in the loss function section.

Upon obtaining M_s, we use it to perform weighted pooling on the visual features V_i' and concatenate the result with the pooled language features L_i' to obtain z_i. The resulting vector is then employed to perform adaptive channel and spatial modulation on the visual features

$$z_i = \mathrm{concat}\left(L_i', \frac{M_s}{\|M_s\|_1 + \epsilon} \cdot V_i'\right). \tag{6}$$

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During channel modulation, z_i is fed into a multilayer perceptron (MLP) to predict channel-wise affine coefficients α and β

$$\alpha, \beta = \mathcal{M}_c(z_i) \tag{7}$$

where $\mathcal{M}_c(\cdot)$ denotes an MLP comprising two fully connected layers, with ReLU activation between layers and Tanh activation at the output. Subsequently, these parameters are applied to adjust the visual features along the channel dimension, yielding the channel-modulated visual features V_i^c

$$V_i^c = LN\left(V_i'\right) \odot \alpha + \beta \tag{8}$$

where α and β are channel-wise coefficients predicted by the MLP and automatically broadcast to the spatial dimensions of V_i. Here, $LN(\cdot)$ denotes layer normalization, and $\odot$ represents element-wise multiplication.

During spatial modulation, z_i is used to predict dynamic convolutional kernel weights W and bias b through another MLP

$$W, b = \mathcal{M}_s(z_i) \tag{9}$$

where $\mathcal{M}_s(\cdot)$ stands for an MLP consisting of a single fully connected layer followed by a Tanh activation function. Following this, the visual features V_i undergo a dynamic convolution parameterized by W and b, yielding the spatially modulated output:

$$V_i^s = \text{Conv}\left(V_i', W, b\right) \tag{10}$$

where the convolution kernel has a size of 1×1. After obtaining the channel-modulated features V_i^c and spatially modulated features V_i^s, they are concatenated along the channel dimension. Then, a 1×1 convolution is applied to project the concatenated features into a desired dimensionality, resulting in the final text-refined representation $\widetilde{V}_i$

$$\widetilde{V}_i = \text{Conv}_{1\times1}\left(\text{Cat}\left(V_i^c, V_i^s\right)\right). \tag{11}$$

Previous works [8], [9], [10], [11] mainly emphasize L2V modulation, with less attention paid to the role of visual context in language refinement. In our Mamba-based framework, the extracted visual features contain rich contextual information, enabling V2L modulation. Consequently, we not only modulate visual features with language cues, but also refine language features with visual representations.

Specifically, the language features L_{i-1} are first linearly projected to match the dimensionality of the visual features $\widetilde{V}_i$, yielding $\widetilde{L}_i = \text{Linear}(L_{i-1}) \in \mathbb{R}^{N\times C_i}$. Cross-attention is then applied, using $\widetilde{L}_i$ as the query and the spatially flattened sequence $\widetilde{V}_i^f = \text{Flatten}(\widetilde{V}_i) \in \mathbb{R}^{H_iW_i\times C_i}$ as the key and value. The resulting presoftmax attention scores lie in $\mathbb{R}^{N\times H_iW_i}$, where N denotes the number of language tokens and H_iW_i denotes the length of the flattened visual sequence at the ith layer. To further incorporate semantic guidance, the semantic map M_s is resized to match the spatial resolution of the visual feature map at the ith layer, then flattened and broadcast across the N language tokens to obtain $\hat{M}_{s,i} = \text{Broadcast}(\text{Flatten}(\text{Resize}(M_s))) \in \mathbb{R}^{N\times H_iW_i}$, where Resize$(\cdot)$ denotes bilinear interpolation. We then define $M_{s,i}^l = \log(\hat{M}_{s,i} + \epsilon)$ and add it to the attention logits as a semantic bias, thereby guiding the attention to semantically relevant visual regions. The resulting formulation is given below

$$L_i = \widetilde{L}_i + \text{Softmax}\left(\frac{\left(\widetilde{L}_iW^q\right)\cdot\left(\widetilde{V}_i^fW^k\right)^T}{\sqrt{d_k}} + M_{s,i}^l\right)\left(\widetilde{V}_i^fW^v\right) \tag{12}$$

where $W^q, W^k \in \mathbb{R}^{C_i\times d_k}$, and $W^v \in \mathbb{R}^{C_i\times C_i}$ are learnable linear projections, the softmax is applied over the spatial dimension, and ϵ is a small constant for numerical stability.

Then, inspired by [39], we perform pooling on L_i and broadcast it along the spatial dimensions to obtain L_i', so as to match the resolution of $\widetilde{V}_i$. We then concatenate L_i' with $\widetilde{V}_i$ and apply a 1×1 convolution to project the concatenated features back to the channel dimension of $\widetilde{V}_i$, resulting in $\widehat{V}_i$

$$\widehat{V}_i = \text{Conv}_{1\times1}\left(\text{Cat}\left(\text{Expand}\left(\text{Pool}\left(L_i\right)\right), \widetilde{V}_i\right)\right) \tag{13}$$

where Expand$(\cdot)$ performs spatial replication to match the spatial resolution of $\widetilde{V}_i$.

Subsequently, a dual-path scan is performed on both the channel and the spatial dimensions of $\widehat{V}_i$. For channel scanning, $\widehat{V}_i$ is first downsampled to a low-resolution space of size $r \times r$ (with $r = 16$) to reduce computational complexity. It is then flattened along the spatial dimension into a 1-D sequence, with the channel dimension becoming the sequence axis. The resulting features are fed into the SSM to capture cross-channel dependencies. After that, we reconstruct the features back to the $r \times r$ spatial structure and upsample them to the original spatial size, performing a residual connection with the original image features. The formula for this process is as follows:

$$\overline{V}_i = U\left(\mathcal{F}^{-1}\left(\text{SSM}\left(\mathcal{F}\left(D\left(\widehat{V}_i\right)\right)\right)\right)\right) + \widehat{V}_i \tag{14}$$

where $D(\cdot)$ denotes downsampling, $U(\cdot)$ refers to upsampling, SSM$(\cdot)$ represents a 1-D selective scan along the channel dimension, $\mathcal{F}(\cdot)$ flattens the spatial dimensions into a 1-D sequence and treats the channel dimension as the sequence axis, and $\mathcal{F}^{-1}(\cdot)$ reconstructs the processed features into an $r \times r$ spatial structure.

For spatial scanning, the visual features are fed into a VSS layer implemented based on the SS2D architecture. The spatial dimensions of the input features are treated as the sequence axis to perform selective state transformation, and a residual connection is employed to obtain the final output visual features V_i

$$V_i = \text{VSS}\left(\overline{V}_i\right) + \overline{V}_i. \tag{15}$$

C. HSR Decoder

The HSR decoder is composed of several VSS blocks and CSR modules stacked alternately, with its detailed architecture shown in Fig. 2. For the multiscale visual features V_i extracted from the mutual modulated encoder, where $i \in \{1, 2, 3, 4\}$, $V_4 \in \mathbb{R}^{C_4\times H_4\times W_4}$ is first upsampled to match the spatial resolution of $V_3 \in \mathbb{R}^{C_3\times H_3\times W_3}$ using bilinear interpolation.

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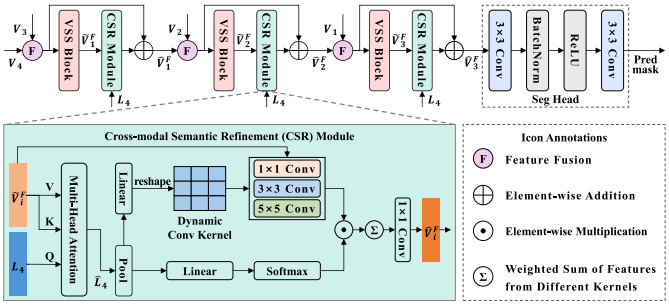


Fig. 2. Structure of the developed HSR decoder.

The spatially aligned features are then concatenated along the channel dimension and further fused through a double-layer convolutional block to produce V_1^{cat}. These operations can be expressed as

$$\begin{aligned} V_1^{cat} &= C_2\left(C_1\left(\mathrm{Cat}\left(U\left(V_4\right), V_3\right)\right)\right) \\ C_i\left(\cdot\right) &= \mathrm{ReLU}\left(BN\left(\mathrm{Conv}_{3\times3}\left(\cdot\right)\right)\right), \quad i \in \{1, 2\} \end{aligned} \tag{16}$$

where $U(\cdot)$ denotes upsampling, $\mathrm{Cat}(\cdot)$ represents concatenation along the channel dimension, and $BN(\cdot)$ indicates batch normalization.

Upon acquiring the fused visual features V_1^{cat}, they are fed into the first VSS block to produce $\widetilde{V}_1^{cat}$. Next, cross-modal attention is computed, where the language features L_4 act as queries and $\widetilde{V}_1^{cat}$ provides the key and value. The corresponding operations are formulated as follows:

$$\begin{aligned} \widetilde{V}_1^{cat} &= \mathrm{VSS}\left(V_1^{cat}\right) \\ \hat{L}_4 &= \mathrm{Softmax}\left(\frac{L_4 W^q \cdot \left(\widetilde{V}_1^{cat} W^k\right)^T}{\sqrt{d_k}}\right) \widetilde{V}_1^{cat} W^v \cdot L_4 \end{aligned} \tag{17}$$

where $\mathrm{VSS}(\cdot)$ denotes the VSS block, and W^q, W^k, and W^v represent the query, key, and value projection matrices, respectively.

Subsequently, multiscale dynamic convolutional kernel weights are derived from the language features $\hat{L}_4$. Specifically, for each kernel size $k_j \in \{1, 3, 5\}$, the kernel weights w_j are computed as a linear transformation of pooled language features. With w_j obtained, multiscale convolutions are applied to the visual features $\widetilde{V}_1^{cat}$. The overall process is mathematically expressed as

$$\begin{aligned} w_j &= \mathrm{Linear}\left(\mathrm{Pool}\left(\hat{L}_4\right)\right) \\ O_j &= \mathrm{Conv}\left(\widetilde{V}_1^{cat}, w_j\right). \end{aligned} \tag{18}$$

Meanwhile, a learnable weighting scheme is applied to the multiscale convolution outputs, where the weights α_j are derived from $\hat{L}_4$ via a linear transformation followed by softmax

$$\alpha_j = \mathrm{Softmax}\left(\mathrm{Linear}\left(\hat{L}_4\right)\right). \tag{19}$$

The outputs from the multiscale convolutions are then fused with weights α_j, followed by a 1×1 convolution to adjust the feature dimensionality. Finally, a residual connection is applied with the visual features V_1^{cat} that have not passed through the VSS block, yielding the output features of the first decoder layer

$$\widehat{V}_1^{cat} = \mathrm{Conv}_{1\times1}\left(\sum_{j=1}^{K} \alpha_j \cdot O_j\right) + V_1^{cat}. \tag{20}$$

Next, $\widehat{V}_1^{cat}$ is concatenated with V_2, and the operations of the first decoder layer described above are repeated to obtain the output of the second decoder layer, $\widehat{V}_2^{cat}$. Following this, $\widehat{V}_2^{cat}$ is concatenated with $\widehat{V}_1$, and the operations of the first and second decoder layers are applied again to produce the output of the third decoder layer, $\widehat{V}_3^{cat}$. Finally, $\widehat{V}_3^{cat}$ is passed through a segmentation head to predict pixel-level masks, as shown in the following formula:

$$\widehat{M} = \mathrm{Conv}_{3\times3}\left(\mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(\widehat{V}_3^{cat}\right)\right)\right)\right). \tag{21}$$

D. Loss Function

Since SK distilled from SAM is incorporated during training, we optimize a total objective comprising segmentation and distillation losses, with the latter computed as the mean squared error (mse) between M_s and M_t

$$\mathcal{L}_{\mathrm{distill}} = \mathrm{MSE}\left(M_s, M_t\right). \tag{22}$$

For the segmentation loss, we follow the loss design adopted in [39] and combine Dice loss [69] and Focal loss [70] with equal weights:

$$\mathcal{L}_{\mathrm{seg}} = \mathcal{L}_{\mathrm{dice}}\left(\widehat{M}, M\right) + \mathcal{L}_{\mathrm{focal}}\left(\widehat{M}, M\right) \tag{23}$$

where $\widehat{M}$ and M denote the predicted mask and the ground-truth annotation, respectively.

The total objective is formulated as

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{seg}} + \lambda \mathcal{L}_{\mathrm{distill}} \tag{24}$$

where λ controls the relative contribution of the distillation term and is analyzed in the model analysis section.

IV. Experiments

A. Datasets and Evaluation Metrics

The proposed method is evaluated on three datasets for RRSIS: RefSegRS [8], RRSIS-D [9], and RISBench [41].

RefSegRS contains 4420 annotated instances over 14 semantic categories, each with five attributes for fine-grained specification. It includes 2172 training, 431 validation, and 1817 test samples. All images are resized to 512×512 pixels at 0.13 m spatial resolution.

RRSIS-D includes 17 402 aligned image–text-mask triplets covering 20 object categories, each with a referring expression enhanced by seven linguistic attributes. Images are resized to 800×800 pixels, with ground sampling distances ranging from 0.5 to 30 m. The dataset comprises 12 181 training, 1740 validation, and 3481 test instances.

RISBench provides 52 472 pixel-annotated image–text-mask samples across 26 categories, each with eight attribute types. It is split into 26 300 training, 10 013 validation, and 16 158 test instances. All images are resized to 512×512 pixels with ground sampling distances from 0.1 to 30 m.

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TABLE I

COMPARISON OF EXPERIMENTAL RESULTS BETWEEN THE PROPOSED METHOD AND SOTA APPROACHES ON THE RRSIS-D DATASET. THE BEST RESULTS FOR EACH METRIC ARE HIGHLIGHTED IN BOLD

Methods	Vision Encoder	Language Encoder	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU
BRINet [59]	R-101	LSTM	56.90	48.77	39.12	27.03	8.73	69.88	49.65
LSCM [60]	R-101	LSTM	56.02	46.25	37.70	25.28	8.27	69.05	49.92
CMPC [61]	R-101	LSTM	55.83	47.40	36.94	25.45	9.19	69.22	49.24
RRN [62]	R-101	LSTM	51.07	42.11	32.74	21.57	6.37	66.43	45.64
CMPC+ [63]	R-101	LSTM	57.65	47.51	36.97	24.33	7.78	68.64	50.24
CRIS [64]	R-101	CLIP	54.84	46.77	38.06	28.15	11.52	70.46	49.69
BKINet [65]	R-101	CLIP	56.90	48.77	39.12	27.03	8.73	69.88	49.65
ETRIS [66]	R-101	CLIP	61.07	50.99	40.94	29.30	11.43	71.06	54.21
CARIS [35]	Swin-B	BERT	71.50	63.52	52.92	40.94	23.90	77.17	62.17
RefSegformer [34]	Swin-B	BERT	66.59	59.58	49.93	33.78	23.30	77.40	58.99
CrossVLT [36]	Swin-B	BERT	66.42	59.41	49.76	38.67	23.30	75.48	58.48
DMMI [37]	Swin-B	BERT	68.74	60.96	50.33	38.38	21.63	76.20	60.12
LAVT [38]	Swin-B	BERT	69.67	63.87	54.14	42.02	24.96	77.66	61.25
LGCE [8]	Swin-B	BERT	69.90	63.96	53.56	40.81	24.21	77.23	60.91
RMSIN [9]	Swin-B	BERT	74.15	67.08	55.64	41.65	23.07	77.49	63.95
LSCF [11]	Swin-B	BERT	74.30	67.69	56.32	43.08	25.67	77.42	64.25
FIANet [40]	Swin-B	BERT	74.46	66.96	56.31	42.83	24.13	76.91	64.01
CroBiM [41]	Swin-B	BERT	75.00	66.32	54.31	41.09	21.78	76.37	64.24
Enti-TwistNet [67]	Swin-B	BERT	77.94	68.81	57.07	44.63	24.80	78.16	66.83
Ours	VMamba-B	BERT	**79.78**	**75.61**	**65.53**	**50.76**	**31.49**	**79.52**	**69.36**

To comprehensively assess performance on RRSIS, we adopt evaluation metrics that are standard in the referring segmentation community, including overall intersection-over-union (oIoU), mean IoU (mIoU), and precision evaluated at the standard thresholds of 0.5, 0.6, 0.7, 0.8, and 0.9, abbreviated as Pr@X.

B. Implementation Details

The language encoder is implemented using BERT from Hugging Face, while the visual encoder is initialized with VMamba [19] pretrained on ImageNet-1K [71]. For distillation, we adopt SAM with a ViT-H backbone (SAM-H) as the frozen teacher, with λ set to 0.4 in the total loss. Optimization is performed with AdamW, applying a learning rate of 2.5×10^{-5} to the visual and language encoders, while the remaining components are updated with 5×10^{-5}. A uniform weight decay of 1×10^{-4} is applied across all parameters. All input images are cropped to a size of 480×480. Experiments are implemented in PyTorch using 4 NVIDIA GeForce RTX 4090 GPUs. The model is trained for 50 epochs on each of the RRSIS-D [9], RISBench [41], and RefSegRS [8] datasets.

C. Comparison With State-of-the-Art Methods

Tables I–III, respectively, present results of the proposed method against state-of-the-art approaches on the RRSIS-D, RISBench, and RefSegRS datasets, with optimal results highlighted in bold. Our method achieves the best overall performance, validating its effectiveness. On the RRSIS-D dataset, the proposed method outperforms the recent method Enti-TwistNet [67] by 1.84%, 1.36%, and 2.53% in terms of Pr@0.5, oIoU, and mIoU, respectively. Similarly, on the RIS-Bench dataset, our method surpasses Enti-TwistNet across all metrics. These experimental results suggest that the proposed Mamba-based architecture is competitive with representative transformer-based models, demonstrating its effectiveness and potential for the RRSIS task.

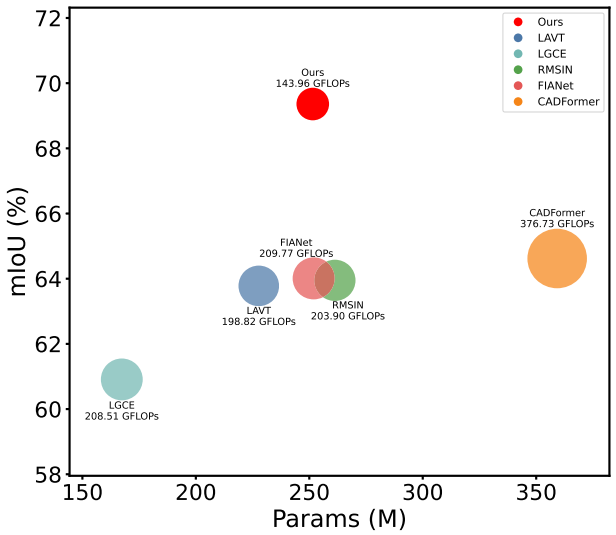


Fig. 3. Comparison with representative transformer-based baselines in terms of parameters, FLOPs, and mIoU. The bubble size indicates FLOPs.

For a fairer comparison with representative transformer-based baselines, we additionally compare parameter count, FLOPs, and mIoU in Fig. 3. As shown in the figure, our method achieves better segmentation performance than

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TABLE II

COMPARISON WITH STATE-OF-THE-ART APPROACHES ON THE RISBENCH DATASET. THE BEST RESULTS FOR EACH METRIC ARE HIGHLIGHTED IN BOLD

Methods	Vision Encoder	Language Encoder	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU
BRINet [59]	R-101	LSTM	52.87	45.39	38.64	30.79	11.86	48.73	42.91
LSCM [60]	R-101	LSTM	55.26	47.14	40.10	33.29	13.91	50.08	43.69
CMPC [61]	R-101	LSTM	55.17	47.84	40.28	32.87	14.55	50.24	43.82
RRN [62]	R-101	LSTM	55.04	47.31	39.86	32.58	13.24	49.67	43.18
MattNet [68]	R-101	LSTM	56.83	48.02	41.75	34.18	15.26	51.24	45.71
CMPC+ [63]	R-101	LSTM	58.02	49.00	42.53	35.26	17.88	53.98	46.73
ETRIS [66]	R-101	CLIP	60.98	51.88	39.87	24.49	11.18	67.61	53.06
CRIS [64]	R-101	CLIP	63.67	55.73	44.42	28.80	13.27	69.11	55.18
CrossVLT [36]	Swin-B	BERT	70.62	65.05	57.40	45.80	26.10	74.33	62.84
CARIS [35]	Swin-B	BERT	73.94	68.93	62.08	50.31	29.08	75.10	65.79
DMMI [37]	Swin-B	BERT	72.05	66.48	59.07	47.16	26.57	74.82	63.93
RefSegformer [34]	Swin-B	BERT	69.15	63.24	55.33	43.27	24.20	74.23	61.25
LAVT [38]	Swin-B	BERT	71.33	66.32	59.37	48.64	28.84	74.71	63.78
RMSIN [9]	Swin-B	BERT	74.54	69.51	62.32	51.53	30.58	73.35	66.86
LGCE [8]	Swin-B	BERT	71.19	65.92	58.94	47.87	27.86	74.34	63.39
LSCF [11]	Swin-B	BERT	76.08	71.29	64.96	55.13	36.73	74.88	68.53
FIANet [40]	Swin-B	BERT	74.80	69.86	63.16	52.70	31.87	74.23	66.83
CroBiM [41]	Swin-B	BERT	75.75	70.34	63.12	51.12	28.45	73.61	67.32
Enti-TwistNet [67]	Swin-B	BERT	76.13	71.47	63.48	52.38	27.46	74.07	67.50
Ours	VMamba-B	BERT	**79.57**	**75.59**	**70.26**	**62.58**	**38.45**	**74.91**	**72.64**

TABLE III

COMPARISON OF THE PROPOSED APPROACH WITH LEADING METHODS ON THE REFSEGRS TEST SET. TOP PERFORMANCE IS HIGHLIGHTED IN BOLD

Methods	Vision Encoder	Language Encoder	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU
BRINet [59]	R-101	LSTM	20.72	14.26	9.87	2.98	1.14	58.22	31.51
LSCM [60]	R-101	LSTM	31.54	20.41	9.51	5.29	0.84	61.27	35.54
CMPC [61]	R-101	LSTM	32.36	14.14	6.55	1.76	0.22	55.39	40.63
CMSA [72]	R-101	None	28.07	20.25	12.71	5.61	0.83	64.53	41.47
RRN [62]	R-101	LSTM	30.26	23.01	14.87	7.17	0.98	65.06	41.88
CMPC+ [63]	R-101	LSTM	49.19	28.31	15.31	8.12	2.55	66.53	43.65
MAttNet [68]	R-101	LSTM	28.79	22.51	11.32	3.62	0.79	64.28	33.42
CRIS [64]	R-101	CLIP	35.77	24.11	14.36	6.38	1.21	65.87	43.26
BKINet [65]	R-101	CLIP	36.12	20.62	15.22	6.26	1.33	63.37	40.41
ETRIS [66]	R-101	CLIP	35.77	23.00	13.98	6.44	1.10	65.96	43.11
CARIS [35]	Swin-B	BERT	45.40	27.19	15.08	8.87	1.98	69.74	42.66
RefSegformer [34]	Swin-B	BERT	50.25	28.01	17.83	9.19	2.48	71.13	47.12
CrossVLT [36]	Swin-B	BERT	41.94	25.43	15.19	3.71	1.76	69.73	42.81
DMMI [37]	Swin-B	BERT	63.89	44.30	19.81	6.49	1.00	68.58	52.15
LAVT [38]	Swin-B	BERT	63.37	48.73	27.17	12.71	3.69	75.04	52.89
LGCE [8]	Swin-B	BERT	74.04	61.39	39.71	17.71	5.17	77.71	59.59
RMSIN [9]	Swin-B	BERT	71.07	57.26	34.87	13.15	2.75	72.68	57.91
CroBiM [41]	Swin-B	BERT	64.83	44.41	17.28	9.69	2.20	72.30	52.69
Enti-TwistNet [67]	Swin-B	BERT	77.72	69.30	45.01	26.66	8.49	78.98	63.82
Ours	VMamba-B	BERT	**82.78**	**78.53**	**55.28**	**47.59**	**16.23**	**80.37**	**72.87**

RMSIN and CADFormer with both fewer parameters and lower FLOPs. Compared with FIANet, our method also attains a higher mIoU at a comparable parameter scale while requiring fewer FLOPs. These results indicate that our method achieves a favorable tradeoff between segmentation performance and computational cost.

Figs. 4–6 intuitively illustrate the visual comparison results between the proposed method and other advanced approaches

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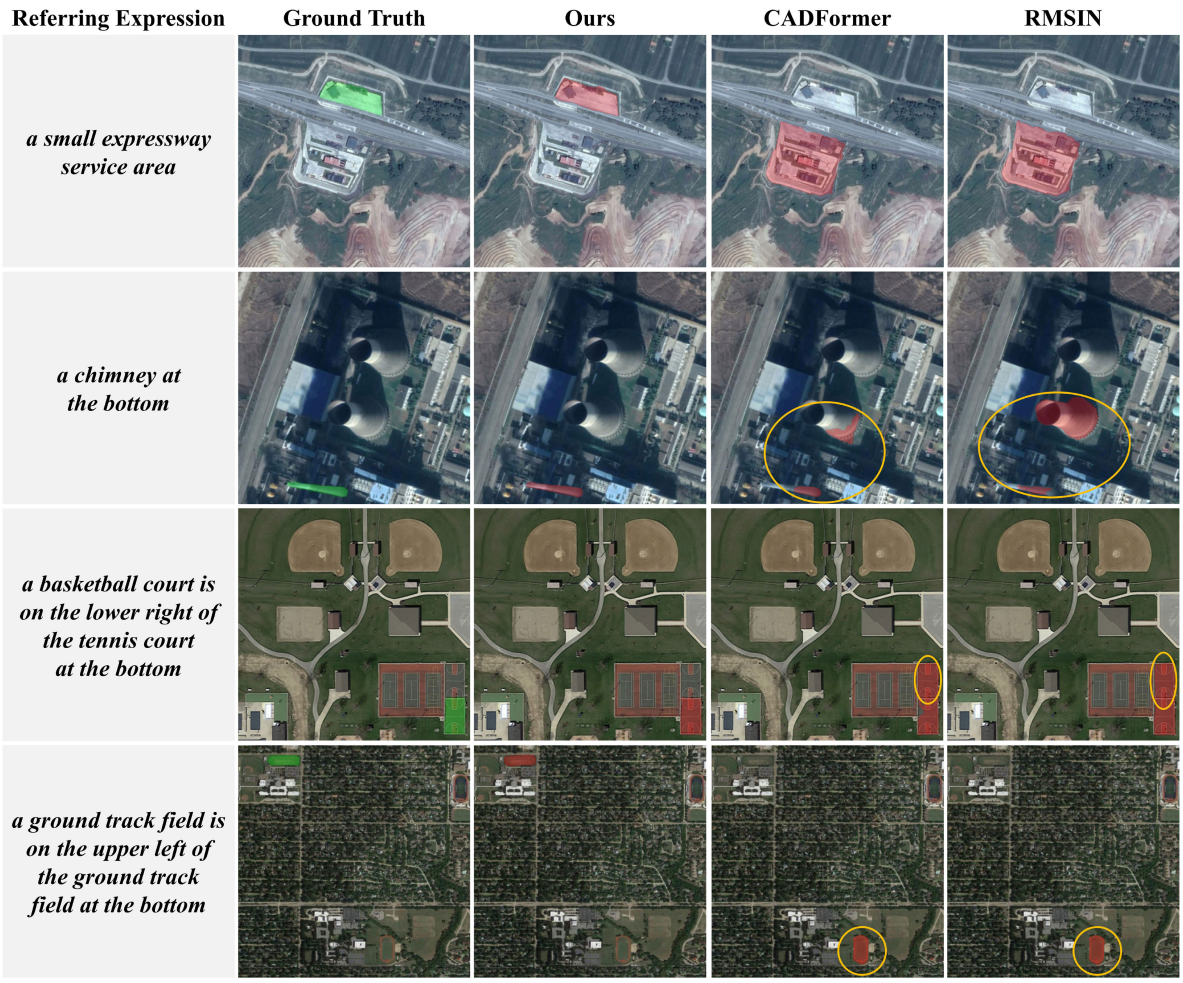


Fig. 4. Visual comparison of different methods on the RRSIS-D dataset. Green masks denote the ground truth; red masks show predictions from the models.

on the three datasets. Even under complex referring expressions, our method accurately segments the target regions, reflecting its superior ability to foster deep visual–language interaction and establish precise cross-modal correspondences. For instance, given the referring expression "a ground track field is on the upper left of the ground track field at the bottom," the proposed model produces an accurate segmentation. In contrast, some other models (e.g., RMSIN and CADFormer) fail to correctly grasp the correspondence between the visual content and the linguistic description, mistakenly segmenting the bottom ground track field as the target.

D. Model Analysis

We conduct comprehensive ablation experiments on the test subset of RRSIS-D [9] to assess the contribution of essential components in our framework.

1) Effectiveness of Various Modules: To comprehensively evaluate the individual and combined contributions of the two modules to the proposed model, we conduct systematic ablation studies by removing the VLMM module, the CSR module, and both modules. The results are summarized in Table IV. The findings indicate that incorporating either VLMM or CSR alone improves model performance.

TABLE IV
ABLATION STUDY OF VLMM AND CSR MODULES

VLMM	CSR	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
×	×	76.52	55.97	23.52	76.93	64.56
✓	×	79.12	63.76	29.46	79.03	68.67
×	✓	78.63	61.45	27.88	78.57	67.41
✓	✓	**79.78**	**65.53**	**31.49**	**79.52**	**69.36**

Specifically, on the basis of the baseline, adding the VLMM module leads to improvements of 2.10%, 4.11%, and 2.60% in oIoU, mIoU, and Pr@0.5, respectively. This gain arises from the strengthened visual–language interaction enabled by VLMM, effectively boosting segmentation performance. When only CSR is integrated, the decoder benefits from the fusion of global semantic modeling and language-guided dynamic modulation, leading to notable performance gains as well. For instance, mIoU increases by 2.85% and Pr@0.5 by 2.11%. The combined use of VLMM and CSR further boosts performance, achieving the best results.

2) Effect of Different Designs of VLMM Module: To gain a deeper understanding of the VLMM module, we explore the performance of several of its variants. Specifically, for

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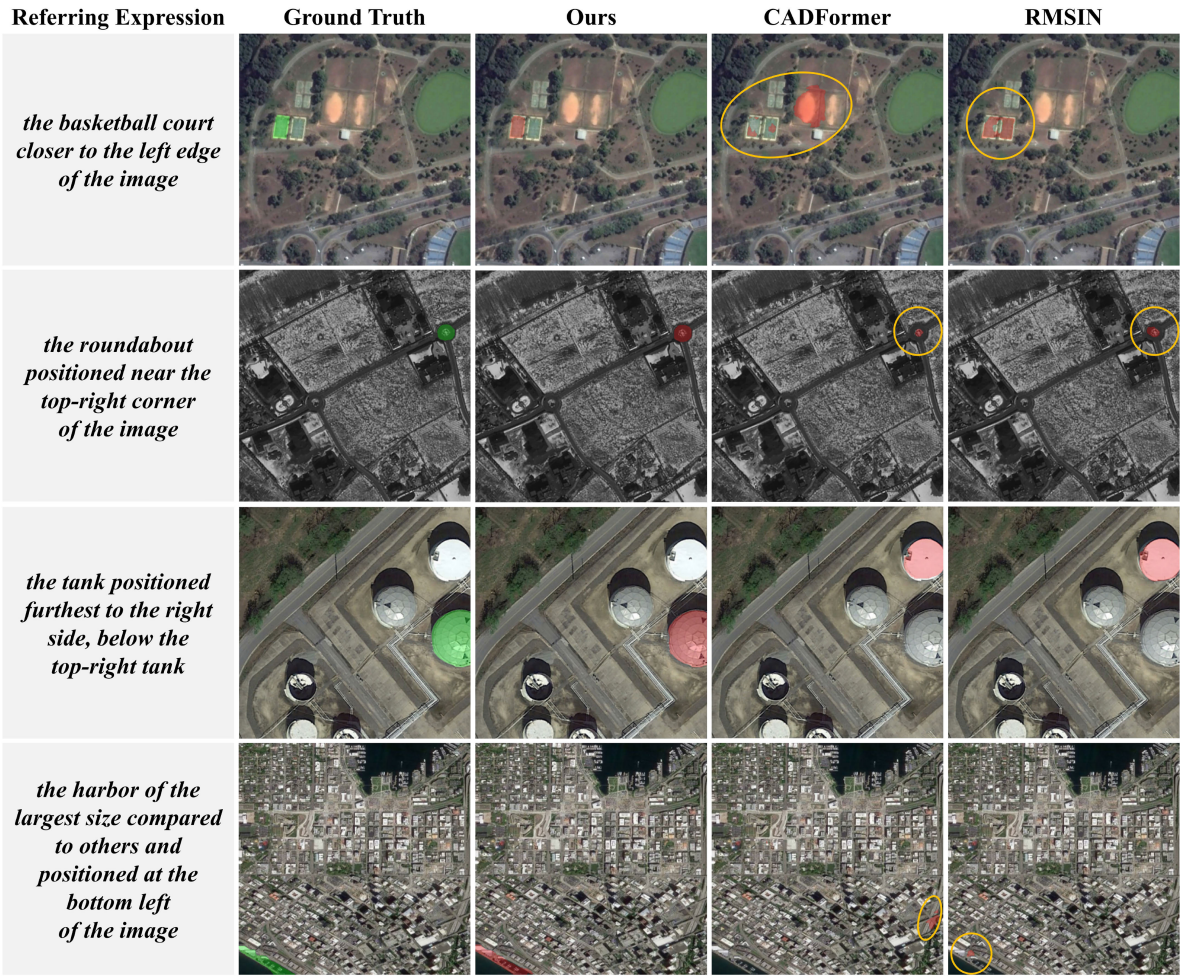


Fig. 5. Visual comparison of different methods on the RISBench dataset. Green masks denote the ground truth; red masks show predictions from the models.

TABLE V

ABLATION STUDY OF CROSS-MODAL INTERACTION STRATEGIES IN THE VLMM DESIGN. L2VONLY REMOVES V2L MODULATION; SYMATTN USES CROSS-ATTENTION FOR BOTH L2V AND V2L

Strategies	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
L2VOnly	78.47	64.02	29.74	78.37	68.05
SymAttn	78.90	64.67	30.45	78.94	68.51
VLMM	**79.78**	**65.53**	**31.49**	**79.52**	**69.36**

TABLE VI

ABLATION STUDY ON L2V MODULATION STRATEGIES IN THE VLMM MODULE. $S \rightarrow C$: SPATIAL BEFORE CHANNEL; $C \rightarrow S$: CHANNEL BEFORE SPATIAL; $S + C$: SUM FUSION; $S \parallel C$: CONCATENATION

Strategies	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
$S \rightarrow C$	77.99	63.34	29.15	78.65	67.89
$C \rightarrow S$	79.45	65.25	31.37	79.25	69.24
$S + C$	79.56	64.98	30.86	79.36	69.16
$S \parallel C$	**79.78**	**65.53**	**31.49**	**79.52**	**69.36**

the VLMM design, we compare three strategies under the guidance of SK: L2VOnly (VLMM without V2L modulation), SymAttn (cross-attention for both L2V and V2L), and our proposed VLMM. As shown in Table V, our method achieves the best performance, demonstrating that bidirectional modulation outperforms standard unidirectional L2V modulation for cross-modal alignment, while symmetric attention alone remains insufficient. In contrast, VLMM enables more precise alignment.

For L2V modulation, we investigate four distinct design paradigms: spatial modulation followed by channel modulation ($S \rightarrow C$), channel modulation followed by spatial modulation ($C \rightarrow S$), the sum of the results of spatial and channel modulation ($C + S$), and the concatenation of both results along the channel dimension ($S \| C$). As shown in Table VI, the concatenation strategy achieves the optimal performance. This is attributed to its preservation of independent information streams from spatial and channel modulation, enabling the complementary fusion of dual pathways. Compared with serial modulation, concatenation avoids information loss caused by the modulation order; in contrast to direct summation, it prevents mutual interference between features and the dilution of response magnitudes, thereby better preserving the integrity of both semantic and spatial information.

For V2L modulation, we introduce a simple correlation-based strategy (CorrMod). It computes cross-modal correlations between visual and language features and performs two forms of weighted pooling over visual features, guided by cross-modal correlations and SK, respectively. The two

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Fig. 6. Visual comparison of different methods on the RefSegRS dataset. Green masks denote the ground truth; red masks show predictions from the models.

TABLE VII

ABLATION STUDY ON VISUAL-TO-LANGUAGE MODULATION STRATEGIES IN THE VLMM MODULE. CORRMOD FOR CORRELATION-BASED MODULATION, MHAMOD FOR ATTENTION-BASED MODULATION

Strategies	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
CorrMod	79.29	64.92	30.66	78.69	68.51
MHAMod	**79.78**	**65.53**	**31.49**	**79.52**	**69.36**

TABLE VIII

ABLATION STUDY OF SK IN DIFFERENT MODULATION BRANCHES

SK in L2V	SK in V2L	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
×	×	78.67	64.32	30.03	78.45	68.19
×	✓	79.38	65.15	31.12	79.27	68.86
✓	×	79.45	65.18	30.96	79.21	68.94
✓	✓	**79.78**	**65.53**	**31.49**	**79.52**	**69.36**

pooled representations are concatenated into a modulation vector, which is then concatenated with language features and passed through a linear layer. The comparison with our multihead attention-based method (MHAMod) is presented in Table VII. MHAMod consistently outperforms CorrMod across all metrics, as it models multiple complementary cross-modal interactions in parallel via multiple attention heads, yielding more accurate language–vision alignment. It achieves a greater performance gain at Pr@0.9 than at Pr@0.5, indicating superior capability in high-precision localization.

3) Effect of SK in Different Branches: To investigate the contribution of SK within VLMM, we conduct ablation studies by removing it from the L2V branch, the V2L branch, or both, so as to isolate its effect in different modulation paths. As shown in Table VIII, integrating SK into either branch yields performance gains over the reference setting without SK, while incorporating it into both branches achieves the best performance. These results suggest that SK is effective in both L2V and V2L modulation and plays complementary roles in the two branches.

4) Disentanglement of Structural and Distillation Gains: To disentangle the effects of the VLMM architecture and distillation, we evaluate three variants: *w/o VLMM*, *VLMM w/o Distillation*, and *VLMM w/ Distillation* (Table IX). The improvement from *w/o VLMM* to *VLMM w/o Distillation* demonstrates the effectiveness of the VLMM design, while the further gains from *VLMM w/ Distillation* indicate the additional benefit of SAM distillation. These results show that the final performance improvement stems from the joint

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TABLE IX

DISENTANGLING THE CONTRIBUTIONS OF VLMM ARCHITECTURE AND SAM DISTILLATION

Model Variants	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
w/o VLMM	78.63	61.45	27.88	78.57	67.41
VLMM w/o Distillation	78.97	64.55	30.46	78.61	68.28
VLMM w/ Distillation	**79.78**	**65.53**	**31.49**	**79.52**	**69.36**

TABLE X

ABLATION STUDY ON THE DISTILLATION LOSS WEIGHT λ

λ	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
0	78.97	64.55	30.46	78.61	68.28
0.2	79.35	65.05	30.87	79.05	68.75
0.4	**79.78**	**65.53**	**31.49**	**79.52**	**69.36**
0.6	79.41	65.15	31.02	79.16	69.05

TABLE XI

COMPARISON OF PERFORMANCE UNDER DIFFERENT SAM TEACHER QUALITIES

Teacher Qualities	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
SAM-B	79.32	65.01	30.98	79.23	68.97
SAM-L	79.51	65.19	31.17	79.31	69.04
SAM-H	**79.78**	**65.53**	**31.49**	**79.52**	**69.36**

contribution of VLMM and SAM distillation, rather than either component alone.

5) Effect of Distillation Loss Weight: We further investigate the influence of the distillation loss weight λ, as summarized in Table X. As λ increases from 0 to 0.4, all metrics improve, with mIoU rising from 68.28% to 69.36%, indicating that a moderate distillation constraint benefits segmentation performance. However, further increasing λ to 0.6 leads to a slight performance drop, suggesting that excessive distillation may interfere with the main segmentation objective. Therefore, we set $\lambda = 0.4$ in the final model.

6) Sensitivity Analysis of SAM Distillation: Beyond the above ablation on the use of SK in VLMM, we further evaluate the distillation scheme from three perspectives, namely teacher quality, bounding-box noise, and alternative teacher models. For consistency, we use the model without distillation reported in Table IX as the reference for comparison.

To investigate the effect of teacher quality, we use three SAM variants, that is, SAM-B, SAM-L, and SAM-H, as teacher models. As shown in Table XI, the overall model performance improves steadily as the teacher becomes stronger, while the gaps among the three variants remain relatively small, suggesting that the proposed distillation scheme is not overly sensitive to teacher quality. Meanwhile, all distilled variants remain better than the model without distillation, indicating that useful supervision can still be transferred even with a weaker teacher.

Moreover, we evaluate the robustness of the framework to noisy prompts by randomly perturbing the bounding boxes, with the results reported in Table XII. Here, $\pm r\%$ denotes simultaneous random jittering of both the box center and box dimensions, with $r \in \{5, 10, 15\}$ in our experiments. As the perturbation level increases, the performance decreases progressively rather than degrading sharply. Even under $\pm 15\%$ noise, the distilled variant still performs better overall than the model without distillation, suggesting that the proposed method is not highly dependent on ideal box prompts.

TABLE XII

COMPARISON UNDER DIFFERENT BOUNDING-BOX NOISE LEVELS

Box Noise Levels	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
None	**79.78**	**65.53**	**31.49**	**79.52**	**69.36**
$\pm 5\%$	79.66	65.39	31.36	79.38	69.25
$\pm 10\%$	79.47	65.07	31.14	79.21	69.08
$\pm 15\%$	79.04	64.83	30.81	78.98	68.65

TABLE XIII

COMPARISON OF DIFFERENT TEACHER MODELS FOR DISTILLATION

Teacher Models	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
OpenSeeD [73]	79.03	64.68	31.25	78.84	68.67
SAM	**79.78**	**65.53**	**31.49**	**79.52**	**69.36**

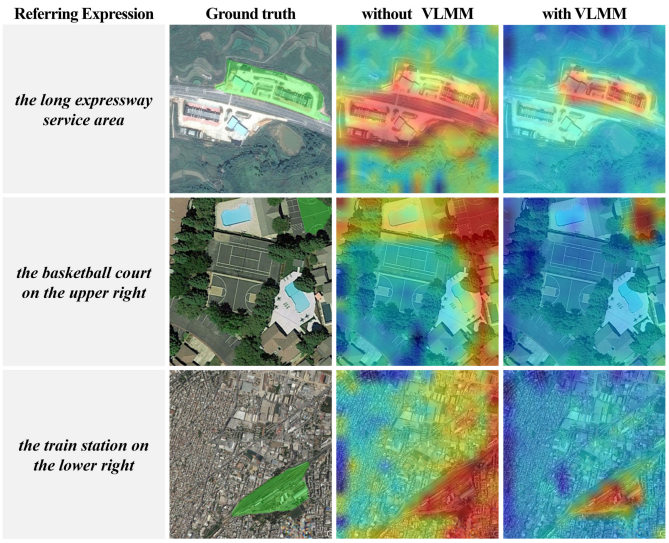


Fig. 7. Visual feature heatmaps of the mutual modulated encoder with and without the VLMM module. Ground truth in green; model predictions in red.

Finally, we replace SAM with OpenSeeD [73] to examine whether the framework can accommodate teachers beyond SAM. As shown in Table XIII, OpenSeeD also improves upon the model without distillation, showing that the proposed framework is not restricted to a single teacher architecture. Nevertheless, SAM yields superior results, suggesting that both teacher quality and the compatibility between the teacher and the task affect the final distillation performance.

7) Qualitative Analysis of VLMM: To intuitively demonstrate the effectiveness of the proposed VLMM module, we visualize the feature maps from the mutual modulated encoder with and without VLMM, with results presented in Fig. 7. With VLMM enabled, feature activations are more precisely localized to language-relevant semantic regions, while activations in irrelevant areas are suppressed. This indicates that VLMM effectively promotes semantic alignment between vision and language, enabling the model to focus

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TABLE XIV
COMPARISON BETWEEN STANDARD CROSS-ATTENTION AND THE PROPOSED CSR IN THE DECODER

Decoder Variants	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
Std. Cross-Attn	78.66	64.08	29.95	79.48	68.23
CSR	**79.78**	**65.53**	**31.49**	**79.52**	**69.36**

TABLE XV
ABLATION STUDY ON CONVOLUTIONAL KERNEL CONFIGURATIONS IN THE CSR MODULE

Conv Kernel Configurations	Pr@0.5	Pr@0.7	Pr@0.9	oIoU	mIoU
1×1	77.61	62.94	28.78	77.74	67.49
$1 \times 1 + 3 \times 3$	78.39	64.48	30.91	78.71	68.61
$1 \times 1 + 3 \times 3 + 5 \times 5$	**79.78**	**65.53**	**31.49**	**79.52**	**69.36**
$1 \times 1 + 3 \times 3 + 5 \times 5 + 7 \times 7$	79.55	65.08	31.12	79.37	68.94

more precisely on visual regions consistent with the language description. For example, given the referring expression "the long expressway service area," the model fails to distinguish between the two service areas in the image without the VLMM module. In contrast, with VLMM, the model exhibits a stronger response in the region corresponding to the longer expressway service area, reflecting its enhanced referring localization capability.

8) Effect of Different Designs of HSR Decoder: To verify the effectiveness and necessity of CSR in the HSR decoder, we construct a variant by replacing CSR with standard cross-attention at all decoding layers while keeping all other settings unchanged. Unlike standard cross-attention, which directly updates visual features, CSR first updates language features and then derives multiscale dynamic convolutional kernels to refine visual features through dynamic filtering. The results are reported in Table XIV. The decoder with CSR consistently outperforms its cross-attention counterpart across all metrics, suggesting that CSR provides more effective visual refinement and thus improves segmentation performance.

We further investigate the effect of different convolutional kernel configurations in CSR. Specifically, we evaluate four variants: 1×1, $1 \times 1 + 3 \times 3$, $1 \times 1 + 3 \times 3 + 5 \times 5$, and $1 \times 1 + 3 \times 3 + 5 \times 5 + 7 \times 7$, as reported in Table XV. The $1 \times 1 + 3 \times 3 + 5 \times 5$ setting achieves the best performance, suggesting that combining small and medium receptive fields may better balance local detail refinement and contextual modeling. Adding the 7×7 kernel leads to a slight performance drop. This may be related to the redundant context introduced by an overly large receptive field, which could weaken local structure modeling. Therefore, we adopt the $1 \times 1 + 3 \times 3 + 5 \times 5$ configuration in CSR.

9) Computational Efficiency Analysis: To assess computational efficiency, we compare our method with representative transformer-based RIS models across different input resolutions in terms of training GPU memory, inference throughput, FLOPs, and parameters (Fig. 8). For fairness, all methods are evaluated on the same NVIDIA GeForce RTX 4090 GPU with a batch size of 1 and without mixed precision. Training memory is measured as the peak usage during one training iteration, while the other metrics are obtained under the same settings. As input resolution increases, competing methods show a steeper rise in memory consumption and reach the GPU memory limit earlier, whereas the memory consumption of our method increases more gradually. Moreover, our advantages in inference throughput and FLOPs become more pronounced at higher resolutions. With a parameter count comparable to LAVT and lower than CADFormer, our method achieves a better tradeoff between performance and computational cost.

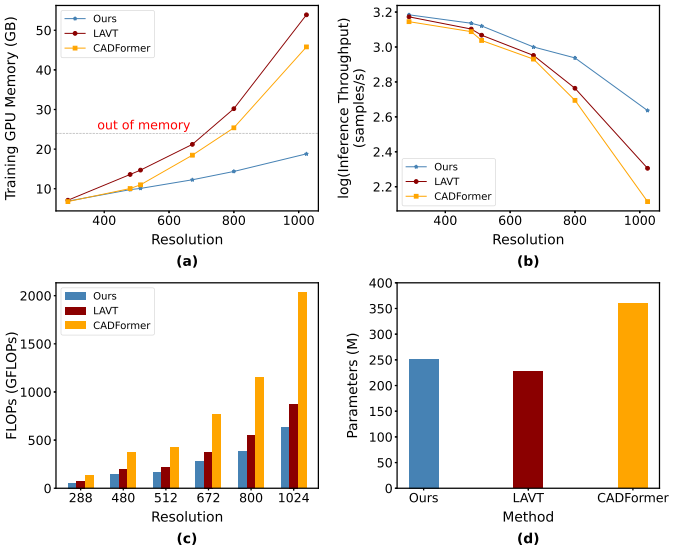


Fig. 8. Efficiency comparison in terms of (a) training GPU memory, (b) inference throughput, (c) FLOPs, and (d) parameter count. All evaluations are conducted on the same NVIDIA GeForce RTX 4090 GPU with a batch size of 1 and without mixed precision.

TABLE XVI
GENERALIZATION RESULTS UNDER UNSEEN-CATEGORY SETTINGS

Methods	Airplane		Vehicle		Airplane + Vehicle	
	Pr@0.5	mIoU	Pr@0.5	mIoU	Pr@0.5	mIoU
LAVT [38]	48.48	41.47	4.48	5.25	11.11	10.71
CADFormer [10]	36.36	40.62	8.06	7.98	12.33	12.90
RMSIN [9]	45.45	42.65	4.84	5.66	10.96	11.23
FIANet [40]	46.46	42.58	3.76	6.18	10.20	11.66
LGCE [8]	37.37	36.53	6.45	6.52	11.11	11.05
Ours	**50.51**	**44.76**	**9.68**	**9.57**	**15.83**	**14.87**

10) Generalization Analysis: We examine the generalization ability of the proposed method from two perspectives: generalization to unseen categories and cross-dataset transfer.

For generalization to unseen categories, two representative categories in remote sensing scenes, *airplane* and *vehicle*, are selected. All training samples of these categories are removed, and the model is evaluated under three settings: *Airplane*, *Vehicle*, and their combination (*Airplane* + *Vehicle*). As reported in Table XVI, our method achieves the best results across all settings, demonstrating that the proposed framework exhibits a degree of generalization to unseen categories. This tendency may stem from the category-agnostic semantic guidance provided by SAM distillation, which could reduce reliance on category-specific appearance patterns.

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TABLE XVII

CROSS-DATASET TRANSFER RESULTS ON RRSIS-D AND RISBENCH. HERE, RRSIS-D → RISBENCH DENOTES TRAINING ON RRSIS-D AND TESTING ON RISBENCH, WHEREAS RISBENCH → RRSIS-D REFERS TO THE REVERSE SETTING

Methods	RRSIS-D → RISBench		RISBench → RRSIS-D	
	Pr@0.5	mIoU	Pr@0.5	mIoU
LAVT [38]	27.74	26.68	39.59	36.48
CADFormer [10]	46.48	43.78	48.00	45.39
RMSIN [9]	38.50	36.15	48.91	53.66
FIANet [40]	40.47	40.69	55.13	50.35
LGCE [8]	34.48	33.45	52.48	47.91
Ours	**49.79**	**46.92**	**58.65**	**56.13**

For cross-dataset transfer, we conduct experiments between RRSIS-D and RISBench by training the model on one dataset and directly testing it on the other without any adaptation. The results in Table XVII show that our method consistently surpasses the compared approaches in both transfer directions. These results indicate that our method remains robust and adaptable under changes in data distribution. This adaptability might also be related to the region-level guidance from SAM distillation, which could alleviate the influence of dataset-specific background and appearance biases.

V. CONCLUSION

This work proposes M^{3}RIS, a Mamba-based framework that investigates the applicability of Mamba in RRSIS. M^{3}RIS leverages the VLMM module to facilitate bidirectional interaction between visual and linguistic features and achieve precise cross-modal semantic alignment, while the HSR decoder jointly captures long-range spatial dependencies and local structural details to enhance semantic awareness of the referred object, thereby producing high-quality segmentation results. The proposed model achieves competitive performance across multiple RRSIS benchmarks, suggesting that the Mamba architecture represents a promising modeling paradigm for vision–language tasks in remote sensing and deserves further exploration. Future efforts will investigate the integration of the Mamba architecture into remote sensing agents to better parse referring expressions and generalize to unseen geographic scenes.

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